Personalized Fashion Recommendation System

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Introduction

Background

- E-Commerce has seen a lot of growth over the past decade. Especially, due to Covid-19 Online Shopping has seen increase in traffic.
- Unlike local markets, online shopping platforms manage comparatively huge amount of goods of different varieties.
- An effective recommender system should be able to help user in effectively navigating the items with ease and suggest products that have higher probability of being bought.

Introduction

Techniques

There exist numerous Recommendation System Models for personalized fashion recommendations. Some the well known ones are:

- Collaborative Filtering
- Content Based Recommendation System
- RNN Recommendation System
- Knowledge Graph:
  - GraphSage
  - Apriory Algorithm
  - PageRank

Introduction

Challenges

- Data size is huge and high dimensional.
- Domain of clothing can be very complex with several factors affecting the user's decision, due to huge variety and quantity of products.
- Cold start problem

Project Aim

- The aim of this project is to build an effective personalized fashion recommendation system which alleviates fore-mentioned challenges, with improved performance compared to other contemporary models.

Related Work

Ups and Downs: Modeling the Visual Evolution of Fashion Trends with One-Class Collaborative Filtering [2]

He et al., 2016



Figure 1: Recommendation trends over time

Related Work

Contd.

- This paper uses One Class Collaborative Filtering.
- Combines high-level visual features extracted from a deep convolutional neural network, users' past feedback, as well as evolving trends within the community.
- Demonstrates how fashion has shifted over the past decade.

Related Work

Aspect-Aware Latent Factor Model: Rating Prediction with Ratings and Reviews[1]

Cheng et al., 2018

- A model that effectively combines reviews and ratings.
- Learns user's and item's latent factors based on ratings, and uses weighted matrix to associate those latent factors.
- The overall rating is computed via a linear combination of the aspect ratings, which are weighted by the corresponding aspect importance.

Related Work

A Linked Data Recommender System using a Neighborhood-based Graph Kernel [3]

Ostuni et al., 2014

- A graph based recommendation approach, which represents items in the knowledge base by their neighborhood graphs.
- Effectively exploits the local neighborhood of items.

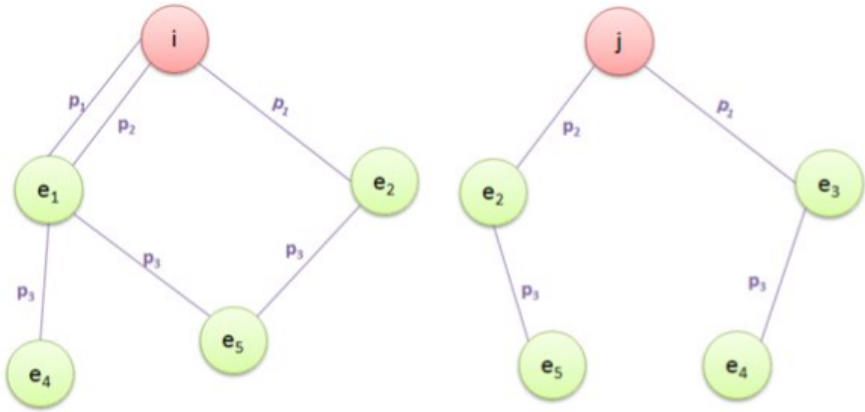


Figure 2: Two-Hop neighborhood traversal

Related Work

Differentiated Fashion Recommendation Using Knowledge Graph and Data Augmentation [5]

Yan et al., 2019

- A different recommendation path for active and inactive users to improve the overall recommendation quality.
- Uses a data augmentation algorithm to label items with tags.
- A user-item knowledge graph is built to discover the potential relationship between items and users.

Related Work

Personalized Clothing Recommendation Based on Knowledge Graph[4]

Wen et al., 2018

- Constructed a knowledge graph of user, a knowledge graph of clothing and a knowledge graph of context.
- Utilized Apriori algorithm to capture correlations between clothing attributes and context attributes.
- *Limitation*: The model does not consider the similarity in different dimensions.

Related Work

Graph Convolutional Neural Networks for Web-Scale Recommender Systems [6]

Ying et al., 2018

- This GCN algorithm combines efficient random walks and graph convolutions to generate embeddings of item nodes.
- Incorporates both graph structure and node feature information.

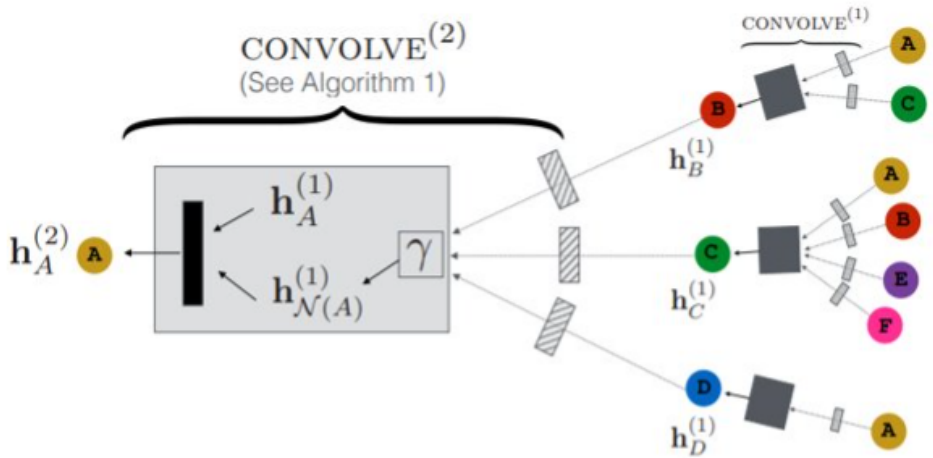


Figure 3: The Graph CNN model- PinSage

Related Work-Summary

	He et al.,	Cheng et al.,	Ostuni et al.,	Yan et al.,	Wen et al.,	Ying et al.,
Image Processing	Yes	No	No	Yes	No	No
Text Processing	Yes	Yes	No	No	No	No
Graph Algorithm	No	No	Yes	Yes	Yes	Yes
Collaborative Filtering	Yes	No	No	No	No	No
Time Sensitive	No	No	No	No	No	No
Sequence Aware	No	No	No	No	No	No

Table 1: **Summary of reviewed papers based on methods used.**

Research Gap

- There is a need for a hybrid recommendation system, since online clothing system is quite complex, there are several factors that come into play which affect the performance of the recommendations.
- For example: Sequential pattern of user's shopping has not been explored.
- Cold Start Problem exists because of data scarcity due to inactive users.
- Real-Time Recommender system is much more adaptable to rapidly changing data, but it hasn't been explored much in fashion recommendation.

Proposed Method

- This project proposes using a hybrid model which is applied on a Knowledge Graph.
- Semantic Analysis to find tags
- Create the Knowledge Graph
- RNN + Collaborative Filtering
- GraphSage Algorithm
- PageRank Algorithm for Cold-Start

Workflow

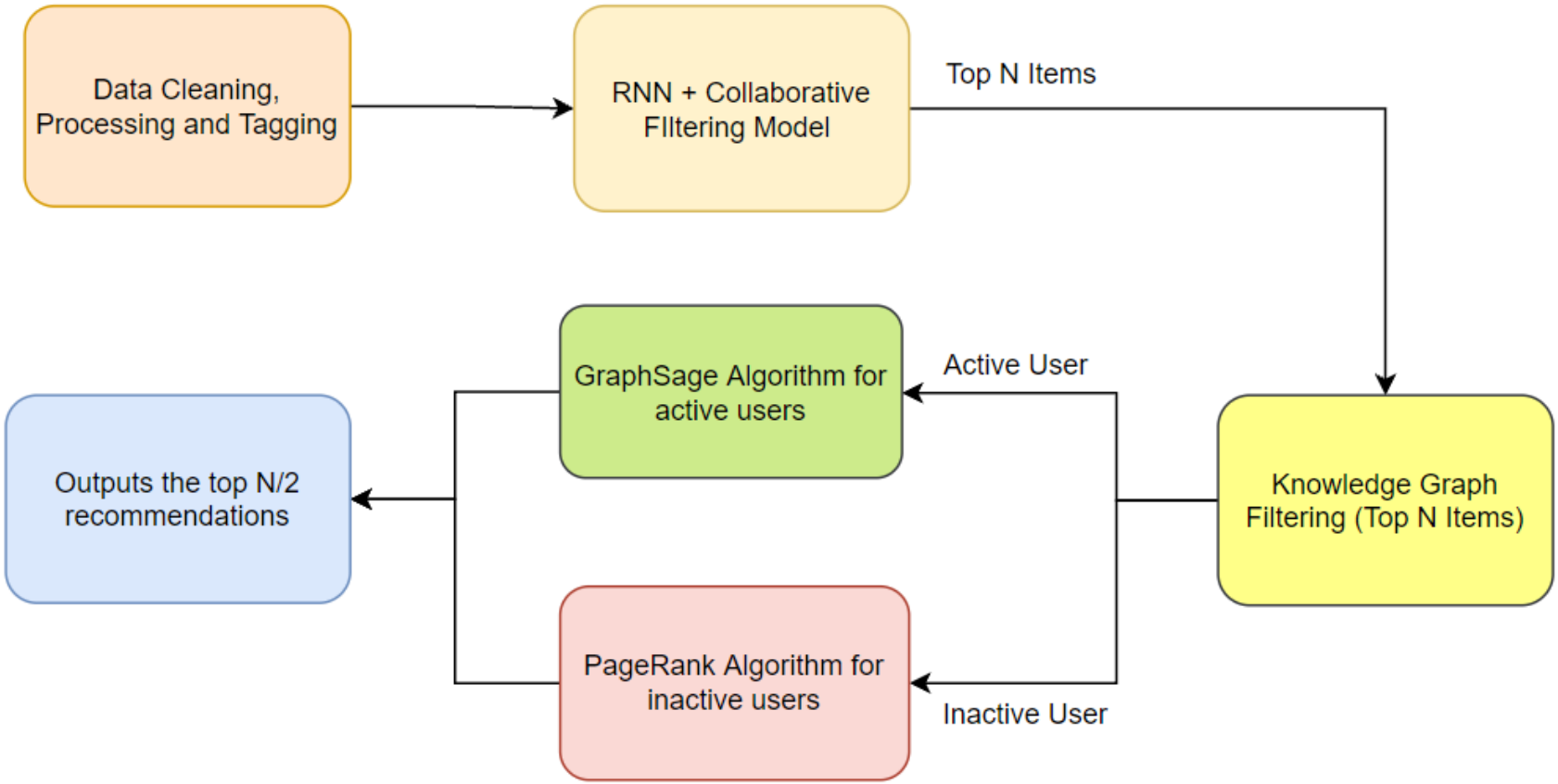


Figure 4: Model Structure

Work Status

- Amazon Review Dataset and Product Metadata, acquired, processed and analysed.
- Collaborative Filtering Implementation.
- Generated the Knowledge Graph.

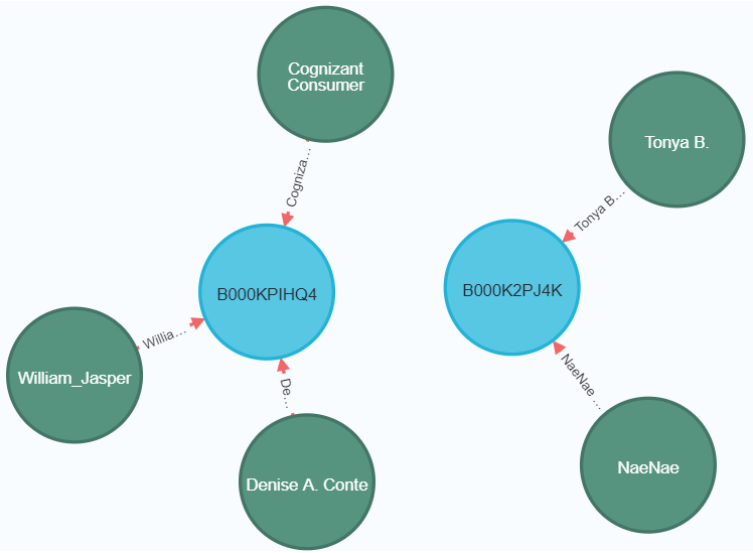


Figure 5: Resulting Knowledge Graph

Validation

- Experimentally we evaluate our method on Amazon Fashion Dataset.
- It's a classic e-commerce dataset, which includes 8,83,000 reviews (ratings, review text, review time, helpfulness votes, 340 Mbs) spanning from May 1996 to July 2014, and item metadata (descriptions, category information, price, brand, and image features, 184 Mbs).
- The Knowledge Graph is built on Neo4j Graph Database Platform.
- Performance of the model will be evaluated based on how accurately the ratings are predicted.

Summary

- We addressed various factors and challenges that could affect a clothing recommendation system.
- We reviewed the existing models that address this problem and their approaches, which helped us identify certain gaps which serve as an opportunity for further improvement.
- We have concluded, there is a need for hybrid recommender systems that addresses the dynamic nature of the domain.
- The implementation of this model would be focused on Amazon Fashion Dataset, which is widely used for this research problem.

Action Plan

- Generating tags for the users and products using Semantics analysis: 7-Oct-2021
- Building RNN model: 14-Oct-2021
- GraphSage Implementation: 21-Oct-2021
- PageRank Implementation: 28-Oct-2021

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